

Critical Values of the t -Statistic.

From the Regularized Incomplete Beta Function to a Practitioner's Reference Table.

AFTER W.S. GOSSET · "STUDENT" · BIOMETRIKA, 1908

§ I The Statistic & Its Pedigree

When the population variance σ^2 is unknown — which it almost always is — the standardised mean of a sample of size n no longer follows the Gaussian distribution law. **Student**, writing in 1908 under a pseudonym imposed by his employer (the Guinness Brewery), worked out the exact distribution of this awkward quantity, and so handed the world its first practical small-sample inferential tool.

The **t-statistic** replaces σ with the sample standard deviation s ; the price is heavier tails, controlled by a single integer parameter ν — the *degrees of freedom*. As $\nu \rightarrow \infty$ the distribution converges, gracefully, to the standard normal distribution.

What follows unpacks this elegant little distribution — its *shape*, its *tails*, and the inference that it quietly powers **more than a century later**.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_\nu, \quad \nu = n - 1 \quad (1)$$

$\bar{x}$
sample mean

μ_0
hypothesised mean under H_0

s
sample standard deviation

ν
degrees of freedom

§ II The Mathematics Beneath the Table

PROBABILITY DENSITY

$$f(x; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad (2)$$

— the t -density, built from the Gamma function —

REGULARISED · INCOMPLETE · BETA

$$\beta(a, b) = \int_0^1 t^{a-1}(1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \quad I(x; a, b) = \frac{\beta(x; a, b)}{\beta(a, b)} \quad (3-4)$$

— Euler's Beta and its regularised cousin on $[0, 1]$, with the symmetry $I(x; a, b) = 1 - I(1-x; b, a)$ —

★ THE BRIDGE TO T DISTRIBUTION

$$F(t; \nu) = \Pr(X \leq t) = 1 - \frac{1}{2} I\left(\frac{\nu}{\nu + t^2}; \frac{\nu}{2}, \frac{1}{2}\right), \quad t > 0 \quad (5)$$

— the t -CDF reduces to one call to the regularised incomplete beta —

LENTZ CONTINUED FRACTION

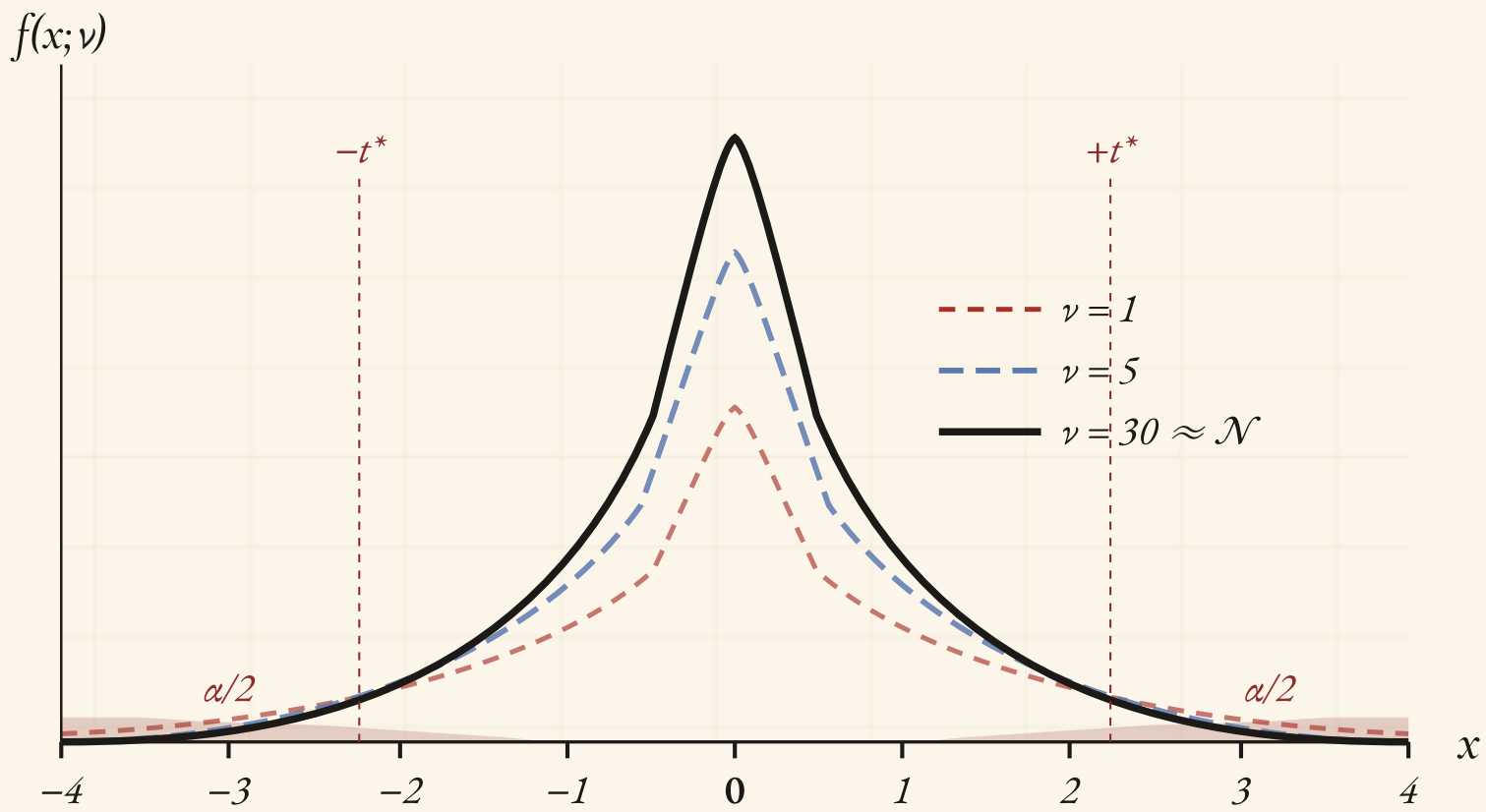
$$I(x; a, b) = \frac{x^a(1-x)^b}{a\beta(a, b)} \cdot \frac{1}{1 + \frac{d_1}{1 + \frac{d_2}{1 + \frac{d_3}{1 + \dots}}}} \quad (6)$$

$$d_{2m} = \frac{m(b-m)x}{(a+2m-1)(a+2m)} \quad | \quad d_{2m+1} = -\frac{(a+m)(a+b+m)x}{(a+2m)(a+2m+1)}$$

INVERSE CDF · NEWTON-RAPHSON METHOD

$$x_{n+1} = x_n - \frac{\text{CDF}(x_n; \nu) - p}{\text{PDF}(x_n; \nu)} \quad (7)$$

— iterate to recover any quantile $t_{\alpha, \nu}$ tabulated below —



From PDF to table. Density (2) is symmetric & bell-shaped, with **tails heavier than the normal**. No closed-form antiderivative exists.

Equation (5) reduces the integral to one call to $I(x; a, b)$, evaluated by the *Lentz continued fraction* (6).

Inverting via Newton-Raphson method (7) recovers any quantile $t_{\alpha, \nu}$ — every entry tabulated below.

§ III Critical Values — *the Table* —

TABLE

Upper-tail Critical Values $t_{\alpha, \nu}$

$\Pr(X > t_{\alpha, \nu}) = \alpha$ | for two-tailed tests use the column $\alpha/2$ | computed from (5)-(7)

ν	significance level α						ν	significance level α					
df	0.10	0.05	0.025	0.01	0.005	0.001	df	0.10	0.05	0.025	0.01	0.005	0.001
1	3.078	6.314	12.706	31.821	63.657	318.31	18	1.330	1.734	2.101	2.552	2.878	3.610
2	1.886	2.920	4.303	6.965	9.925	22.327	20	1.325	1.725	2.086	2.528	2.845	3.552
3	1.638	2.353	3.182	4.541	5.841	10.215	22	1.321	1.717	2.074	2.508	2.819	3.505
4	1.533	2.132	2.776	3.747	4.604	7.173	25	1.316	1.708	2.060	2.485	2.787	3.450
5	1.476	2.015	2.571	3.365	4.032	5.893	28	1.313	1.701	2.048	2.467	2.763	3.408
6	1.440	1.943	2.447	3.143	3.707	5.208	30	1.310	1.697	2.042	2.457	2.750	3.385
7	1.415	1.895	2.365	2.998	3.499	4.785	40	1.303	1.684	2.021	2.423	2.704	3.307
8	1.397	1.860	2.306	2.896	3.355	4.501	50	1.299	1.676	2.009	2.403	2.678	3.261
9	1.383	1.833	2.262	2.821	3.250	4.297	60	1.296	1.671	2.000	2.390	2.660	3.232
10	1.372	1.812	2.228	2.764	3.169	4.144	100	1.290	1.660	1.984	2.364	2.626	3.174
12	1.356	1.782	2.179	2.681	3.055	3.930	120	1.289	1.658	1.980	2.358	2.617	3.160
15	1.341	1.753	2.131	2.602	2.947	3.733	∞	1.282	1.645	1.960	2.326	2.576	3.090

Entries via the inverse regularised incomplete beta $I^{-1}\left(x; \frac{\nu}{2}, \frac{1}{2}\right)$ per (5)-(7).

■ e.g. $t_{0.025, 10} = 2.228$ — two-tailed $\alpha = 0.05$ with $n = 11$.

"Every experiment may be said to exist only in order to give the facts a chance of disproving the null hypothesis."

— R. A. Fisher, *The Design of Experiments*, 1935 —



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